

The code for `$a_2$` is:

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided $\{0.4, 0.6\}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?

Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

The code defines a_2 as the limit of $2f(n) + \frac{1}{2} + 0.25^n$. Evaluating $f(n)$ shows it converges to $\frac{1}{2} + \delta$, where δ is an extremely small positive number (on the order

converges to $\frac{1}{4} + o$, where o is an astronomically small positive number (on the order of $72^{-10^{24}}$). Hence $a_2 = 1 + 2\delta > 1$.

The optimisation problem is to minimise $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$ and $x_1, x_2 \geq 0$. Because $a_2 > 1$, the unique minimiser is $x_1^* = 0, x_2^* = 1/a_2 \approx 1$. The infinity norm distance from the given point $\{0.4, 0.6\}$ to $(0, 1/a_2)$ is $\max(0.4, |0.6 - 1/a_2|)$. Since $1/a_2 \approx 1$, the second term is approximately 0.4, making the distance exactly 0.4 (or microscopically less). This is strictly less than 0.45.